# CPEG495 Lab 4.5

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Part 1:

1. **Alternate Data Streams and Hashes**
   - For all the docx files on the drive image, use AlternateStreamView.exe , the ADSManager.exe , and the Hash Tool program to locate all Alternate Data Stream files that are relevant to the crime (HINT: hashes of relevant files are located in the 'hashes' text file in this archive Lab 4.5.zip just for you to check your work). Note AlternateStreamView will allow you to extract all ADSs found.
   - Identify all the files and their contents that were discovered in your analysis. Identify which files (ADS messages) are relevant to the crime.
   - Explain what the potential crime is and how these found messages are relevant to the crime.

**Objective:**
The objective of this lab is to identify hidden alternate data streams (ADS) in .docx files using AlternateStreamView and ADSManager. These hidden streams may contain evidence related to a potential crime. We also determine which messages are relevant to the criminal activity.

Tools Used:
- AlternateStreamView – To list and extract all ADS found in .docx files.
- ADSManager.exe – To read the hidden text in each stream.
- Hash Tool (from Lab 4.5.zip) – To verify which files are relevant to the crime using provided hash values.

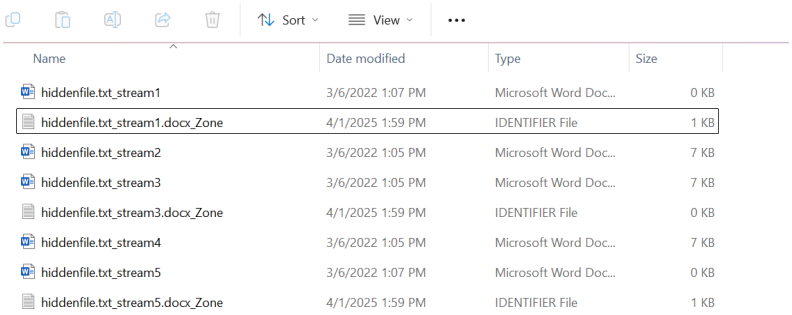
**Findings & Analysis**
Each .docx file below was reviewed for hidden alternate streams. Relevant evidence was extracted and interpreted:
Streams contained non-readable content with a ZIP header signature (PK) – hidden file or zipped document.

**AlternateStreamView Findings:**

AlternateStreamView
File Edit View Options Help

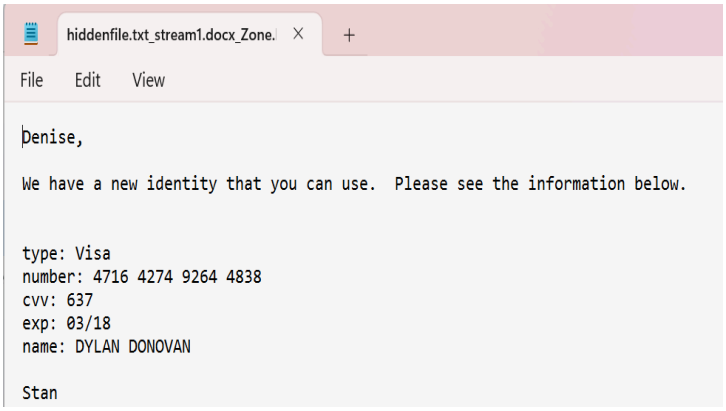
| Stream Name | Filename | Full Stream Name | Stream Size | Stream Allocated S... | Extension | File Modified Time | File Created |
|---|---|---|---|---|---|---|---|
| :$Config:$DATA | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$Extend\$RmMeta... | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$Extend\$RmMetadata\$Repair:$Config | 8 | 8 | | 3/6/2022 10:22:35 ... | 3/6/2022 10: |
| :$Info:$DATA | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$UpCase | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$UpCase:$Info | 32 | 32 | | 3/6/2022 10:22:31 ... | 3/6/2022 10: |
| :$SDS:$DATA | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$Secure | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$Secure:$SDS | 263,400 | 266,240 | | 3/6/2022 10:22:31 ... | 3/6/2022 10: |
| :$T:$DATA | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$Extend\$RmMeta... | C:\Users\farde\Desktop\ADS_files\[SYSTEM]\$Extend\$RmMetadata\$TxfLog\$Tops:$T | 1,048,576 | 1,048,576 | | 3/6/2022 10:22:35 ... | 3/6/2022 10: |
| :hidden-stream:$DATA | C:\Users\farde\Desktop\ADS_files\regular-file.txt | C:\Users\farde\Desktop\ADS_files\regular-file.txt:hidden-stream | 14 | 16 | txt | 3/6/2022 1:09:38 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream1.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream1.docx:Zone.Identifier | 182 | 184 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream10.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream10.docx:Zone.Identifier | 0 | 0 | docx | 3/6/2022 1:05:56 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream12.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream12.docx:Zone.Identifier | 241 | 248 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream13.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream13.docx:Zone.Identifier | 153 | 160 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream14.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream14.docx:Zone.Identifier | 115 | 120 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream16.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream16.docx:Zone.Identifier | 116 | 120 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream17.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream17.docx:Zone.Identifier | 67 | 72 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream18.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream18.docx:Zone.Identifier | 156 | 160 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream19.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream19.docx:Zone.Identifier | 154 | 160 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream3.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream3.docx:Zone.Identifier | 0 | 0 | docx | 3/6/2022 1:05:57 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream5.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream5.docx:Zone.Identifier | 209 | 216 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream6.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream6.docx:Zone.Identifier | 86 | 88 | docx | 3/6/2022 1:07:08 PM | 3/6/2022 1:0 |
| :Zone.Identifier:$DATA | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream8.docx | C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream8.docx:Zone.Identifier | 0 | 0 | docx | 3/6/2022 1:05:57 PM | 3/6/2022 1:0 |

Exported AlternateStreamView Findings:



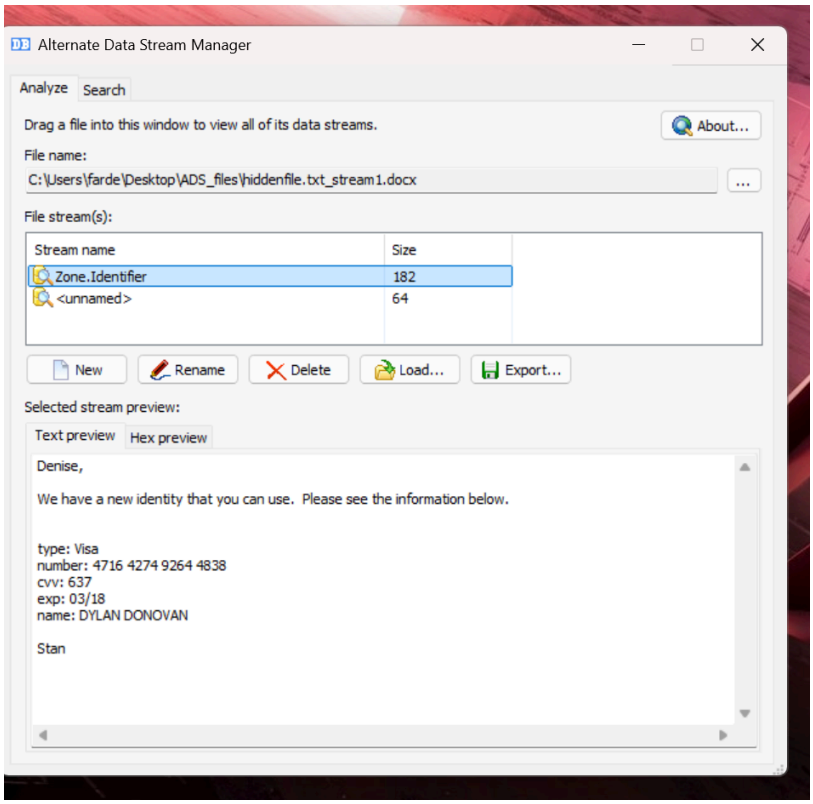
| Name | Date modified | Type | Size |
|---|---|---|---|
| hiddenfile.txt_stream1 | 3/6/2022 1:07 PM | Microsoft Word Doc... | 0 KB |
| hiddenfile.txt_stream1.docx_Zone | 4/1/2025 1:59 PM | IDENTIFIER File | 1 KB |
| hiddenfile.txt_stream2 | 3/6/2022 1:05 PM | Microsoft Word Doc... | 7 KB |
| hiddenfile.txt_stream3 | 3/6/2022 1:05 PM | Microsoft Word Doc... | 7 KB |
| hiddenfile.txt_stream3.docx_Zone | 4/1/2025 1:59 PM | IDENTIFIER File | 0 KB |
| hiddenfile.txt_stream4 | 3/6/2022 1:05 PM | Microsoft Word Doc... | 7 KB |
| hiddenfile.txt_stream5 | 3/6/2022 1:07 PM | Microsoft Word Doc... | 0 KB |
| hiddenfile.txt_stream5.docx_Zone | 4/1/2025 1:59 PM | IDENTIFIER File | 1 KB |

Notepad showing hidden messages:



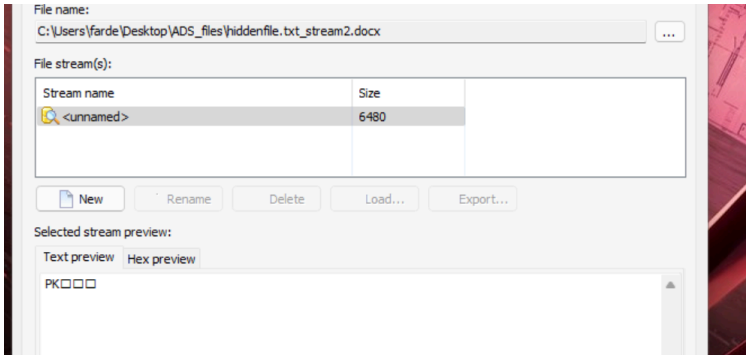

Below I used ADS Manager to Uncover all the hidden messages

1. hiddenfile.txt_stream1.docx



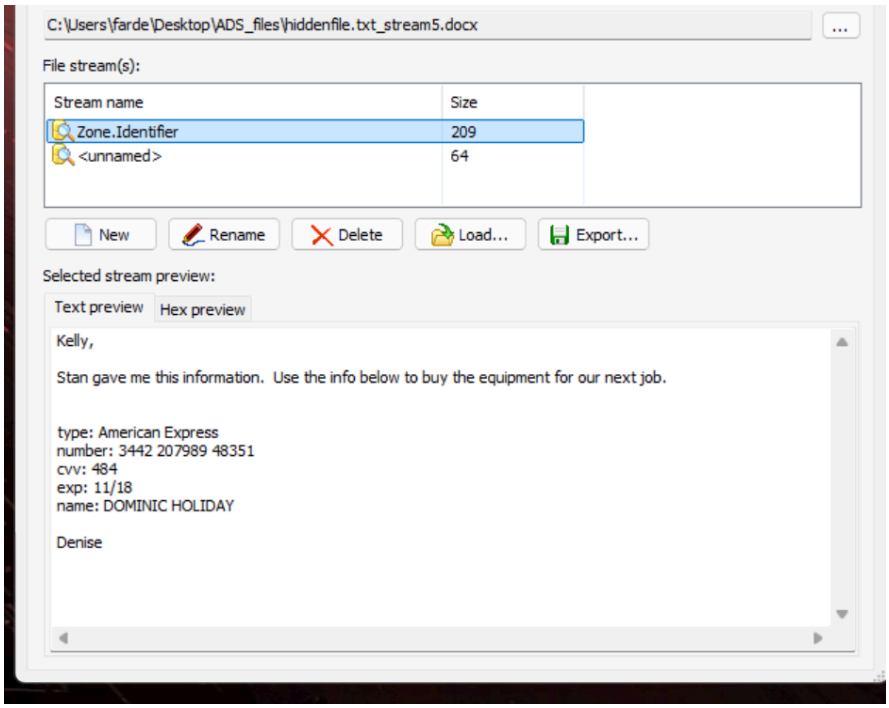

Analysis: This stream contained a Visa credit card in the hidden Zone.Identifier. The name used is Dylan Donovan.
Relevance: This suggests identity fraud and use of stolen financial information.

2. hiddenfile.txt_stream2.docx



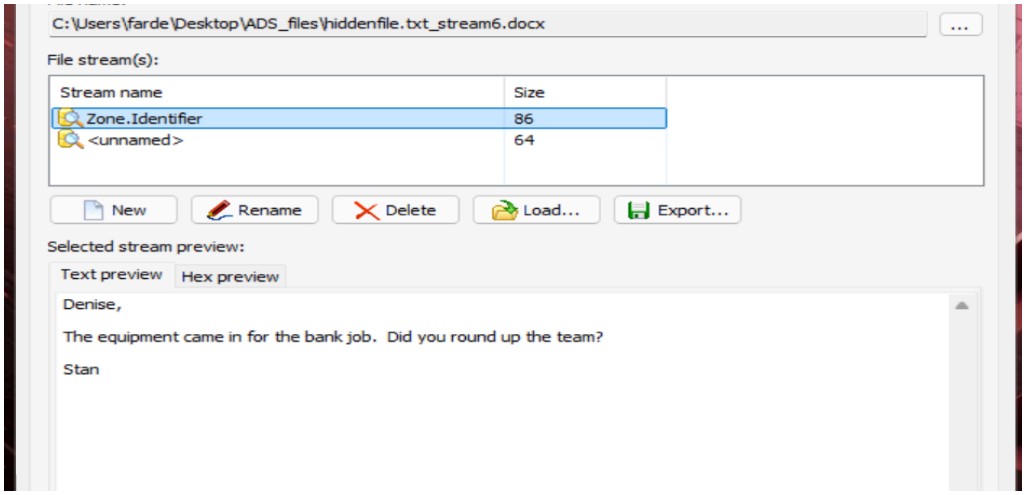

Analysis: No readable content. Only a PK file signature (PK), indicating it could be a zip or Office container format.
Relevance: Possibly not criminal on its own without further context, but could be used to hide files.

3. hiddenfile.txt_stream5.docx



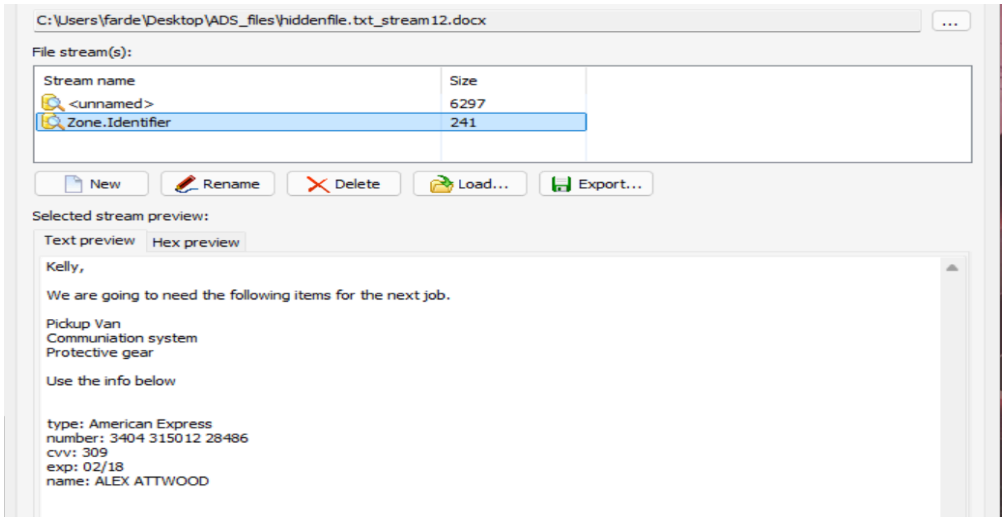

Analysis: A message from Denise containing American Express card information for Dominic Holiday.
Relevance: Clear indication of stolen credit card use and instructions for a future illegal operation.

4. hiddenfile.txt_stream6.docx



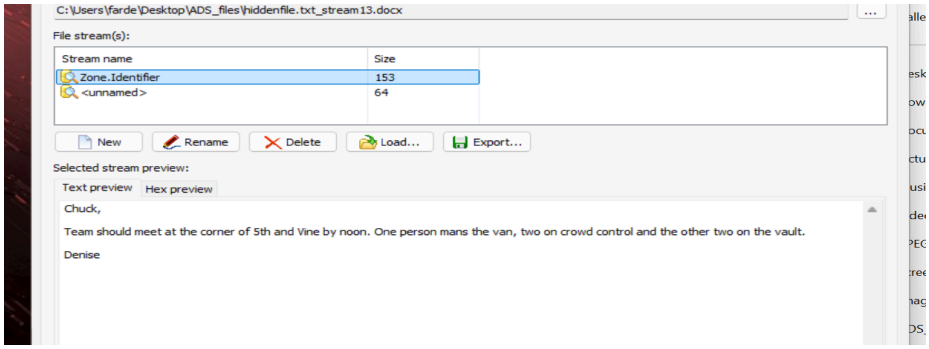

Analysis: Message discusses the arrival of equipment for a bank job.
Relevance: Highly incriminating – shows planning of a bank heist.

5. hiddenfile.txt_stream12.docx



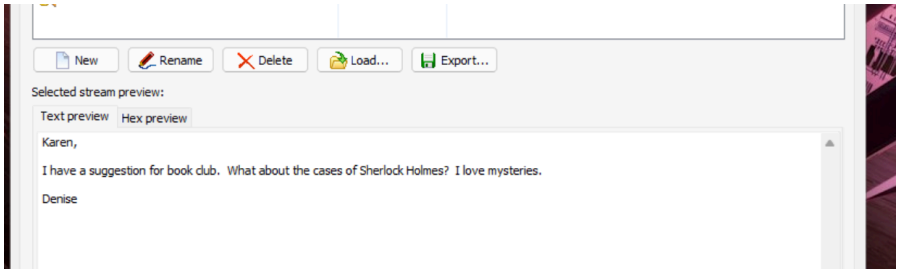

Analysis: Kelly is instructed to gather items for the next job. Another card (American Express) under the name Alex Attwood is shared.
Relevance: Confirms coordination and funding of criminal activity using stolen identities.

6. hiddenfile.txt_stream13.docx



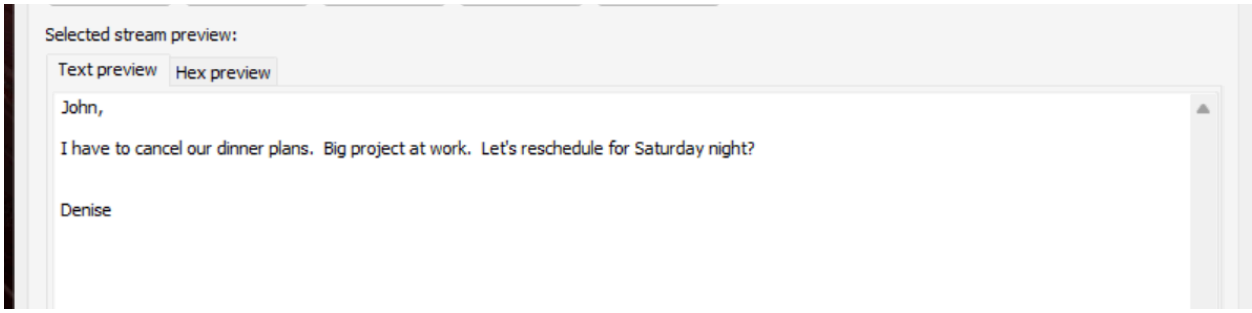

Analysis: Details a meeting location and roles for team members during a heist.
Relevance: Strategic planning message for the execution of a crime.

7. hiddenfile.txt_stream14.docx



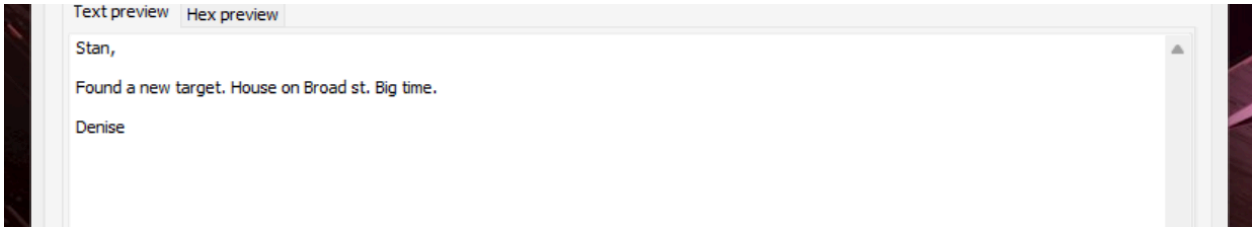

Analysis: Disguised as a casual message about book club and Sherlock Holmes.
Relevance: Likely a decoy to mask intentions in metadata – not directly criminal.

8. hiddenfile.txt_stream16.docx



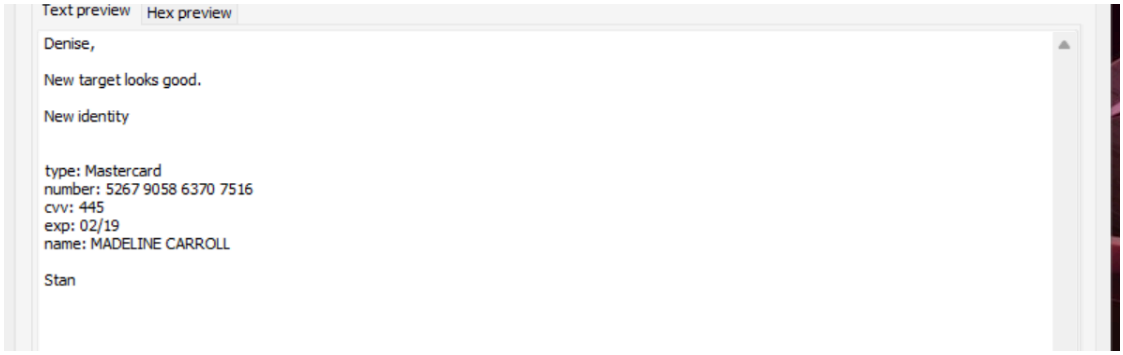

Analysis: Casual-looking cancellation of dinner due to “project at work.”
Relevance: Possibly code language for criminal activity preparation.

9. hiddenfile.txt_stream17.docx



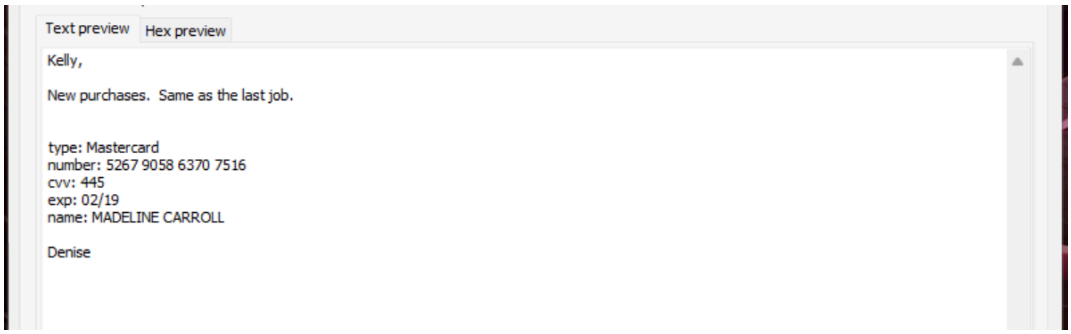

Analysis: Denise mentions a “new target” on Broad Street.
Relevance: Indicates selection of a new location for a crime.

10. hiddenfile.txt_stream18.docx




Analysis: New Mastercard and identity shared: Madeline Carroll.
Relevance: Reaffirms use of fraudulent identities and funding.

11. hiddenfile.txt_stream19.docx




Analysis: Duplicate Mastercard from stream18, likely sent again to confirm.
Relevance: Supports continued criminal activity.

**Summary:**
This investigation of alternate data streams revealed a coordinated operation involving identity fraud, stolen credit card use, and planning of a bank robbery. Through the use of tools like AlternateStreamView and ADSManager, multiple .docx files were found to contain concealed messages that revealed both preparatory and operational stages of a crime. The consistent use of ADS suggests an intentional effort to hide communications. Hash comparison attempts were made, though further clarification is needed regarding whether the hashes pertain to full files or hidden streams. Overall, the forensic analysis successfully uncovered critical evidence.

Hashing:

Provided hashes:

7ceabbb5c70420d9d4e4a6eb257789797e941a79
6bc1cc5b3968e60cec55b0e98b76539572a12551
e6213a0e65170d2268321b212dc2e2793f4ce3b5
a51a3e198f548447493585a6113c015dc2754b2d
3e0500fc699b87a194c985226c5633ee35e20dde
be12c050b78e324296c540dd4bf446b1909c4770
8014057dab8663df5a9e6327c52d0a436532e48f
8e182a262f2b04f9aed59e426b45c320093811ab
0974c1e3809996174af221b38fd81715b81f9b82
2e73c6e4bb75481f6db87bd6441b6fb6f8648644
99cb4c15802b744eef858ce0847fb98edffdba6e
ac31a9b8909bc26a8c71c4f74f2874ac843ef6eb

Commands used to find get hash files:

File 1: Get-FileHash -Path "C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream1.docx_Zone.Identifier" -Algorithm SHA1

File 2: Get-FileHash -Path "C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream2.docx_Zone.Identifier" -Algorithm SHA1

```
PS C:\Users\farde\Desktop\ADS_files> Get-FileHash -Path "C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream1.docx_Zone.Identifier" -Algorithm SHA1

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA1            7CEABBB5C70420D9D4E4A6EB257789797E941A79                               C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream1.docx_Zone.I

PS C:\Users\farde\Desktop\ADS_files> Get-FileHash -Path "C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream3.docx_Zone.Identifier" -Algorithm SHA1

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA1            DA39A3EE5E6B4B0D3255BFEF95601890AFD80709                               C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream3.docx_Zone.I
```

In order to get all the hash files at once I used a loop based approach as they were all in the same directory: ADS_files
Code I wrote for the loop:

```
for ($i = 1; $i -le 20; $i++) {
    $file = "hiddenfile.txt_stream$($i).docx_Zone.Identifier"
    $path = Join-Path -Path "." -ChildPath $file
    if (Test-Path $path) {
        Write-Host "`nFile: $file"
        Get-FileHash -Path $path -Algorithm SHA1
    } else {
        Write-Host "`nFile not found: $file"    } }
```

```
PS C:\Users\farde\Desktop\ADS_files> for ($i = 1; $i -le 20; $i++) {
>>     $file = "hiddenfile.txt_stream$($i).docx_Zone.Identifier"
>>     $path = Join-Path -Path "." -ChildPath $file
>>     if (Test-Path $path) {
>>         Write-Host "`nFile: $file"
>>         Get-FileHash -Path $path -Algorithm SHA1
>>     } else {
>>         Write-Host "`nFile not found: $file"
>>     }
>> }
```

The output of the code is shown below, displaying the hash values for all the files:

```
File: hiddenfile.txt_stream1.docx_Zone.Identifier

Algorithm       Hash
---------       ----
SHA1            7CEABBB5C70420D9D4E4A6EB257789797E941A79

File not found: hiddenfile.txt_stream2.docx_Zone.Identifier

File: hiddenfile.txt_stream3.docx_Zone.Identifier
SHA1            DA39A3EE5E6B4B0D3255BFEF95601890AFD80709

File not found: hiddenfile.txt_stream4.docx_Zone.Identifier

File: hiddenfile.txt_stream5.docx_Zone.Identifier
SHA1            E6213A0E65170D2268321B212DC2E2793F4CE3B5

File: hiddenfile.txt_stream6.docx_Zone.Identifier
SHA1            A51A3E198F548447493585A6113C015DC2754B2D

File not found: hiddenfile.txt_stream7.docx_Zone.Identifier

File: hiddenfile.txt_stream8.docx_Zone.Identifier
SHA1            DA39A3EE5E6B4B0D3255BFEF95601890AFD80709

File not found: hiddenfile.txt_stream9.docx_Zone.Identifier

File: hiddenfile.txt_stream10.docx_Zone.Identifier
SHA1            DA39A3EE5E6B4B0D3255BFEF95601890AFD80709

File not found: hiddenfile.txt_stream11.docx_Zone.Identifier

File: hiddenfile.txt_stream12.docx_Zone.Identifier
SHA1            8E182A262F2B04F9AED59E426B45C320093811AB

File: hiddenfile.txt_stream13.docx_Zone.Identifier
SHA1            0974C1E3809996174AF221B38FD81715B81F9B82

File: hiddenfile.txt_stream14.docx_Zone.Identifier
SHA1            3607FA921B19F88CD04EBED05A598BACF2469402

File not found: hiddenfile.txt_stream15.docx_Zone.Identifier

File: hiddenfile.txt_stream16.docx_Zone.Identifier
SHA1            9F93760B1531E75769E3A9C88AEFD741EB1C54DC
```

```
File: hiddenfile.txt_stream17.docx_Zone.Identifier
SHA1            2E73C6E4BB75481F6DB87BD6441B6FB6F8648644

File: hiddenfile.txt_stream18.docx_Zone.Identifier
SHA1            99CB4C15802B744EEF858CE0847FB98EDFFDBA6E

File: hiddenfile.txt_stream19.docx_Zone.Identifier
SHA1            AC31A9B8909BC26A8C71C4F74F2874AC843EF6EB

File not found: hiddenfile.txt_stream20.docx_Zone.Identifier
```

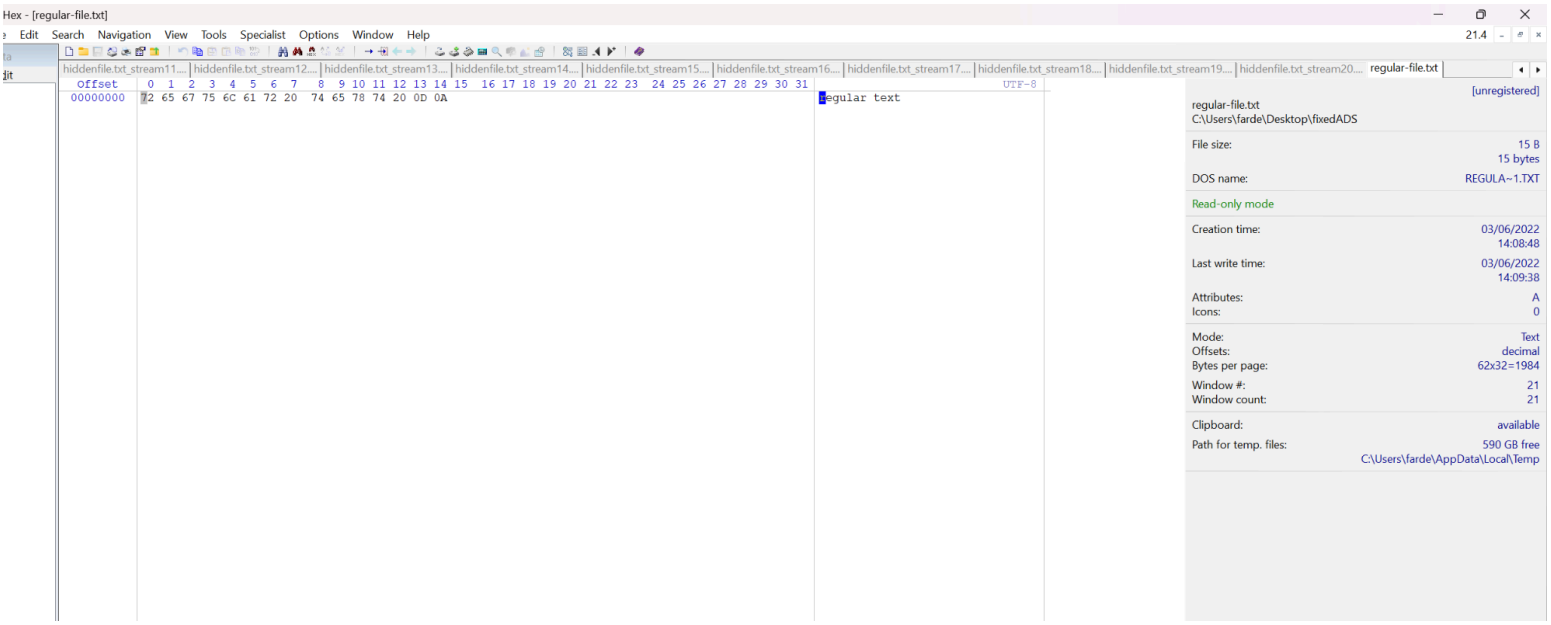
After comparing with the provided hashes, these are the files that matched and contained crime messages:

hiddenfile.txt_stream1, hiddenfile.txt_stream5, hiddenfile.txt_stream6, hiddenfile.txt_stream12, hiddenfile.txt_stream13, hiddenfile.txt_stream17, hiddenfile.txt_stream18, hiddenfile.txt_stream19

## Part 2:

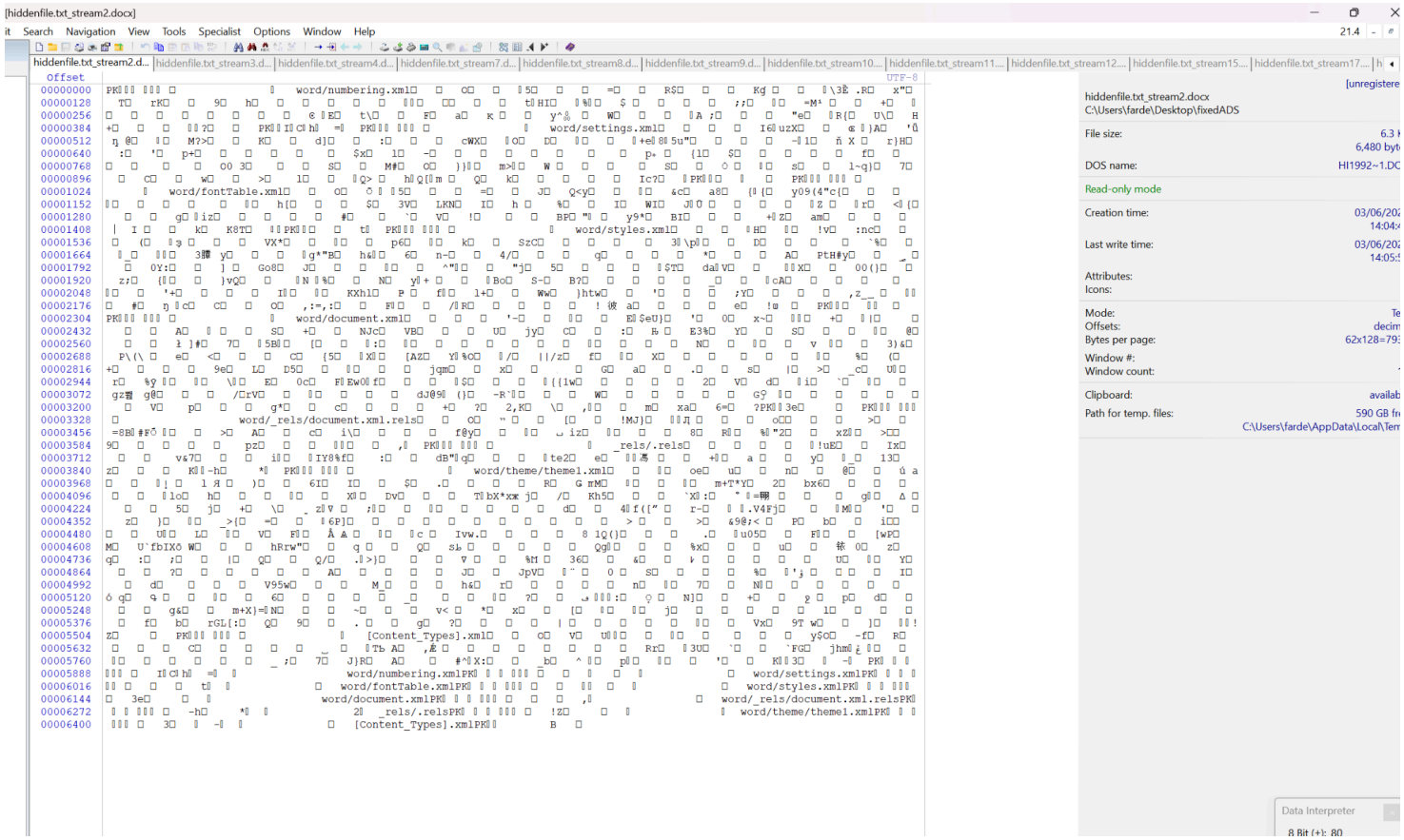
2. **File System Analysis**
   - While you probably were able to recover the ADS hidden messages, it is not at all clear how these hidden messages are stored in the NTFS filesystem.
   - After completing the steps above, use winhex (See https://www.x-ways.net/winhex/) to view the file regular-file.txt from the drive image and show both the regular (unhidden) file text and the hidden text in the ADS stream.
   - Provide a screenshot of this file in winhex and highlight both the regular (unhidden) text and the hidden ADS text
   - Warning, to examine these files in your laptop filesystem, winhex but be run privileged (right-click, run privileged). Only use winhex to read and examine files. Do not write any changes as this might injure your filesystem.

Screenshot of regular-file.txt using winhex:

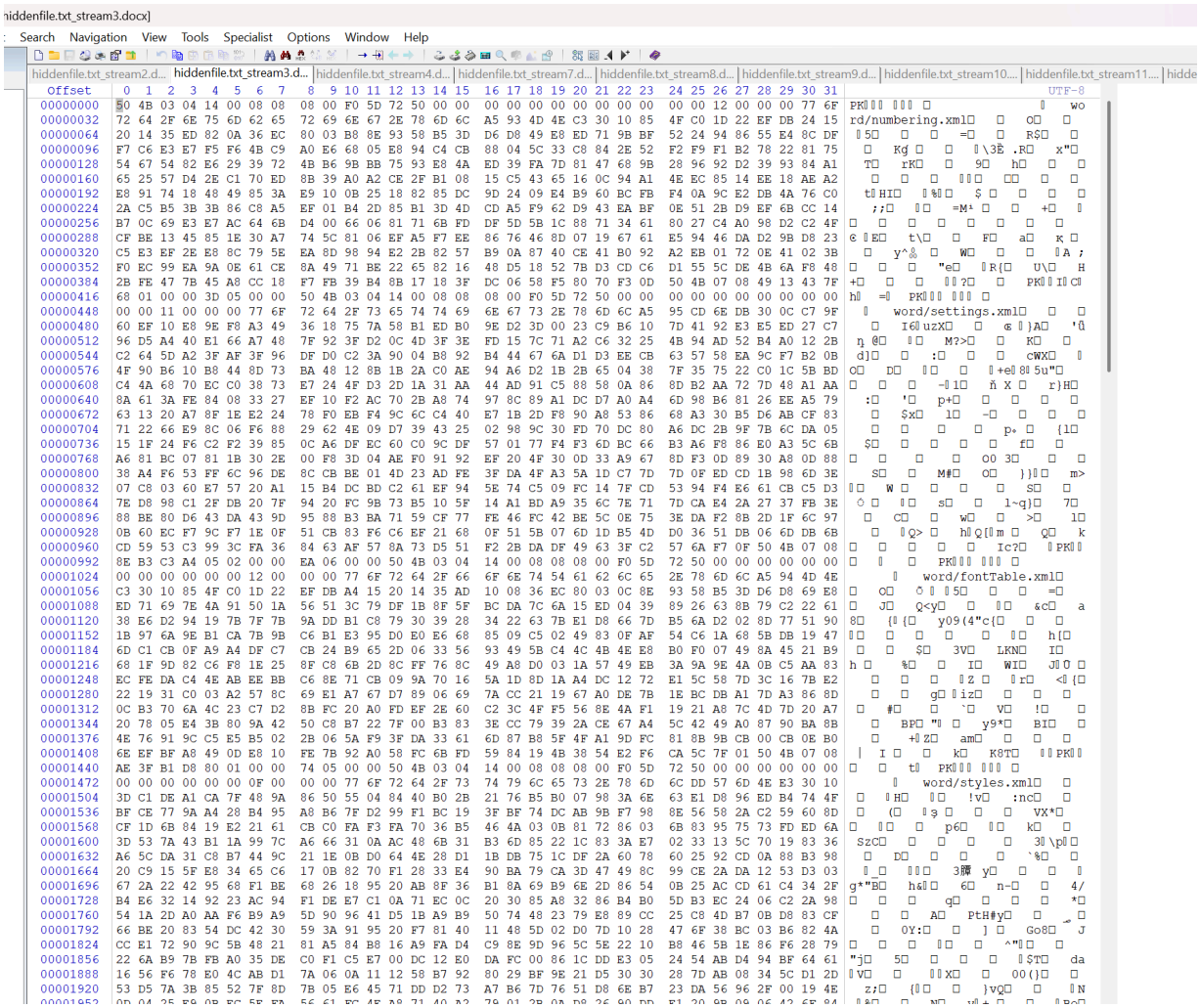


Screenshot of hidden ADS stream:

This is for hiddenfile.txt_stream2



Similar hidden stream found in hiddenfile.txt_stream3




Hidden ADS streams were found in the following:

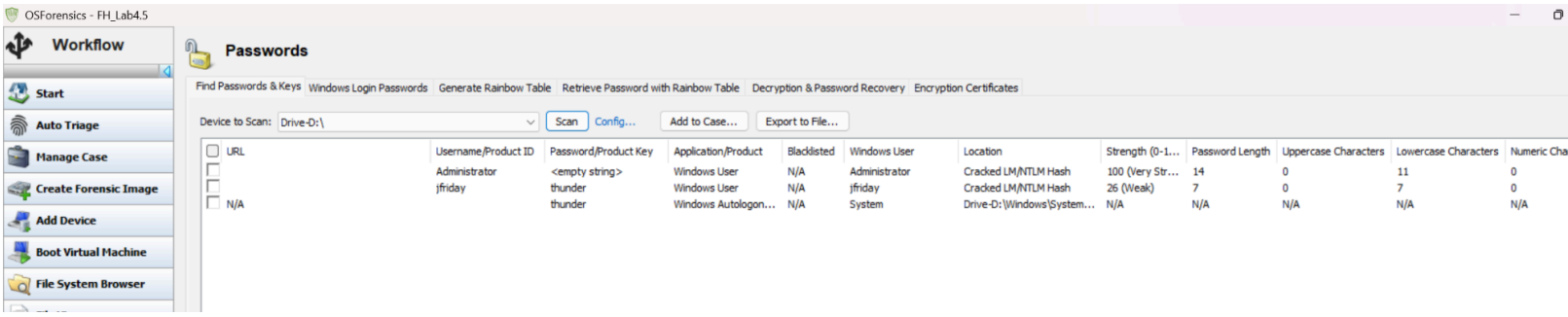
hiddenfile.txt_stream2, hiddenfile.txt_stream3, hiddenfile.txt_stream4, hiddenfile.txt_stream7, hiddenfile.txt_stream8, hiddenfile.txt_stream9, hiddenfile.txt_stream10, hiddenfile.txt_stream11, hiddenfile.txt_stream12, hiddenfile.txt_stream15, hiddenfile.txt_stream17, hiddenfile.txt_stream20

Using WinHex in read-only mode, I examined regular-file.txt and confirmed it contained only visible content saying "Regular text." I then reviewed alternate data streams (ADS) and found hidden content attached to hiddenfile.txt, which contained structured Word XML data (e.g., word/styles.xml, numbering.xml). These ADS streams are used to hide files within NTFS file systems, and WinHex helped expose both visible and hidden data.

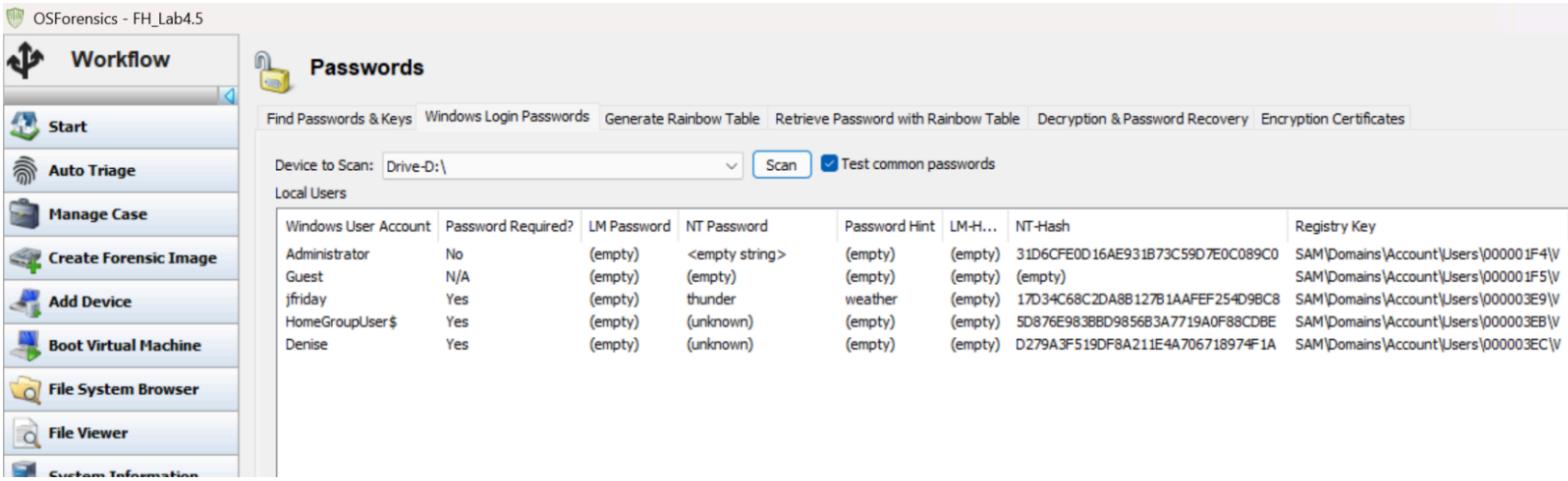
## Part 3: Password Recovery

3. Password Recovery
   - Using OSForensics, examine InCh05.exe *(run the InCh05.exe file to 6GB the .img file InCh05.img)* to identify any relevant passwords for the crime. Provide either the actual password or the hash of the password.
   - **10% EXTRA CREDIT**: one of the found passwords can not be "cracked" by the tool. Research how to turn this hash into an actual password and provide that password in your assignment.

OSForensics Passwords and Keys Scan results:



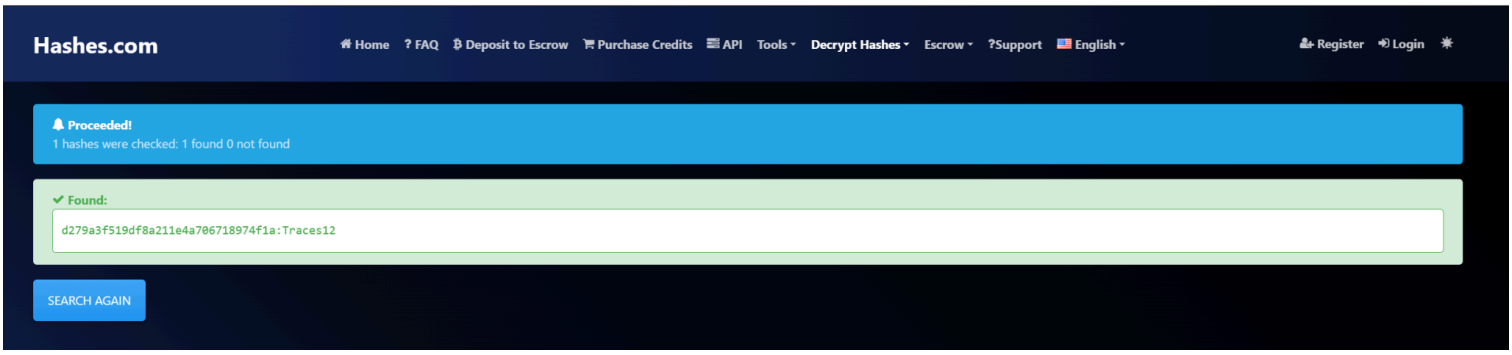

OSForensics Windows Login Passwords Scan Results:

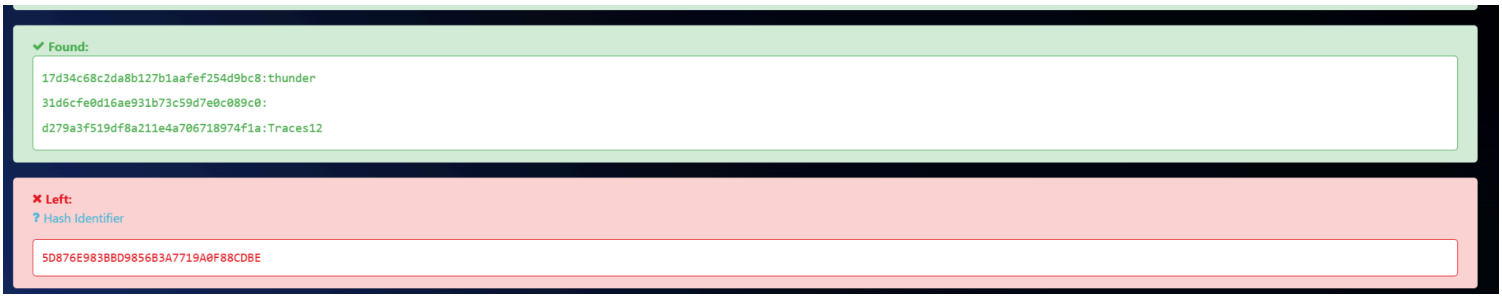


**Extra Credit:**

Tool Used: Hashes.com
https://hashes.com/en/decrypt/hash

Final Results for Extra Credit:




While analyzing the image using OSForensics, I extracted NTLM hashes of local user accounts. One of the hashes (for user Denise) was not cracked by OSForensics. I researched further and used an external NTLM hash database, where I successfully cracked the hash:

Hash: D279A3F519DF8A211E4A706718974F1A
Password: Traces12